A_sub commutator closure step 1 — {Q̂, P̂_op} = 0 on Wallach K-type basis

Lyra (Claude Opus 4.7)

2026-05-26 Tuesday EDT (~08:25 EDT via `date`)

A_sub commutator closure step 1 — {Q̂, P̂_op} = 0

1. The commutator computation

1.1 Operators on Wallach K-type basis

The Wallach K-type basis V_(m_1, m_2) of Bergman $H^2(D\_IV^5)$ carries definite (m_1, m_2) labels under K = SO(5) × SO(2) (or Spin(5) × Pin(2) on the universal cover). On this basis:

Charge operator (T2470, STRUCTURALLY VERIFIED): Q̂ V_(m_1, m_2) = q_K · V_(m_1, m_2), q_K := m_2 (in units of fundamental U(1)_em weight)

Q̂ is the SO(2) weight operator; its eigenvalue on V_K is the m_2 index. Charge is U(1)_em weight, derived from the SO(2) factor of K (T2478 RIGOROUSLY CLOSED).

Parity operator (T2472, STRUCTURALLY VERIFIED): P̂_op = $\hat{\gamma}^5 \circ \sigma$

where σ is the Möbius involution on D_IV^5 acting in Hua coordinates as z → -z̄ on the SO(2) factor, and $\hat{\gamma}^5$ is the Pin(2) Z_2 lift (T2471 RATIFIED). On Wallach K-types:

σ V_(m_1, m_2) = (-1)^{m_2} · V_(m_1, -m_2)

The (-1)^{m_2} arises because the Möbius involution acts on the SO(2) factor with phase factor depending on m_2; the V_(m_1, -m_2) result reflects the σ-induced m_2-sign-flip.

$\hat{\gamma}^5$ V_(m_1, m_2) = ε_K · V_(m_1, m_2), ε_K = +1 (integer m_1, m_2 — bosons), -1 (half-integer m_1, m_2 — fermions)

$\hat{\gamma}^5$ has Z_2 eigenvalue determined by the Pin(2) Z_2 grading (integer sublattice vs half-integer sublattice on the Pin(2) cover, per A_sub v0.4 Section 2.2 partition).

Combining: P̂*op V*(m_1, m_2) = ε_K · (-1)^{m_2} · V_(m_1, -m_2)

1.2 $[\hat{Q}, \hat{P}_{op}]$ explicit computation

Apply $\hat{Q} \circ \hat{P}_{op}$: $\hat{Q} \circ \hat{P}_{op} V(m_1, m_2) = \hat{Q} \cdot \varepsilon_K \cdot (-1)^{m_2} \cdot V(m_1, -m_2) = \varepsilon_K \cdot (-1)^{m_2} \cdot q_{\sigma K} \cdot V_{(m_1, -m_2)} = \varepsilon_K \cdot (-1)^{m_2} \cdot (-m_2) \cdot V_{(m_1, -m_2)}$

(Since $q_{\sigma K} = -m_2$ — the σ-flipped K-type has charge $-m_2$.)

Apply $\hat{P}_{op} \circ \hat{Q}$: $\hat{P}_{op} \circ \hat{Q}\, V(m_1, m_2) = \hat{P}_{op} \cdot m_2 \cdot V(m_1, m_2) = m_2 \cdot \varepsilon_K \cdot (-1)^{m_2} \cdot V(m_1, -m_2)$

Compute the commutator: $[\hat{Q}, \hat{P}_{op}]\, V_{(m_1, m_2)} = (\hat{Q} \circ \hat{P}_{op} - \hat{P}_{op} \circ \hat{Q})\, V(m_1, m_2) = \varepsilon_K \cdot (-1)^{m_2} \cdot (-m_2 - m_2) \cdot V(m_1, -m_2) = -2\, m_2 \cdot \varepsilon_K \cdot (-1)^{m_2} \cdot V_{(m_1, -m_2)} = -2\, m_2 \cdot \hat{P}_{op} V(m_1, m_2) = -2\, \hat{Q}\, V_{(m_1, m_2)} \cdot \hat{P}_{op}$ (formal)

Equivalently, the ANTI-COMMUTATOR is: $\{\hat{Q}, \hat{P}_{op}\}\, V(m_1, m_2) = (\hat{Q} \circ \hat{P}_{op} + \hat{P}_{op} \circ \hat{Q})\, V(m_1, m_2) = \varepsilon_K \cdot (-1)^{m_2} \cdot (-m_2 + m_2) \cdot V(m_1, -m_2) = 0$

1.3 The structural identity

$\{\hat{Q}, \hat{P}_{op}\} = 0$ on $H^2(D_{IV}^5)$.

Equivalently: **$[\hat{Q}, \hat{P}_{op}] = -2\, \hat{Q} \cdot \hat{P}_{op}$** (or $-2\, \hat{P}_{op} \cdot \hat{Q}$, equivalent expressions).

This is the load-bearing structural identity of the A_{sub} algebra at the Q-P junction.

2. Physical interpretation

2.1 Substrate-parity contains charge-flipping content

The Möbius involution σ on the SO(2) factor negates the m_2 index, hence negates the charge eigenvalue. Combined with the Pin(2) Z_2 lift $\hat{\gamma}^5$, the substrate-level parity operator $\hat{P}_{op}$ effectively flips both spatial orientation (via σ on Hua coordinates) AND charge sign (since charge is the SO(2) weight that σ negates).

This is substantially different from SM parity P (which preserves charge — electron remains electron under P, only mirror-image). The substrate-level parity $\hat{P}_{op}$ is structurally more like substrate-CP than substrate-P alone.

2.2 Connection to T2476 $\alpha^{\text{BST primary}}$ substrate-mechanism

T2476 (Friday 2026-05-22) established that the QED fine-structure constant α has an exponent pattern $\alpha^{\text{BST primary}}$ reflecting substrate structure. The $\{\hat{Q}, \hat{P}_{op}\} = 0$ anti-commutation is the operator-algebra source of this pattern:

- $\hat{Q}$ generates $U(1)_{em}$ gauge transformations
- $\hat{P}_{op}$ acts as substrate-level discrete symmetry (including charge-flip content)
- Their anti-commutation forces specific structure in QED amplitudes when both are present

In QED loop expansion, the anti-commutation contributes to anomalous dimension structure. The α^{BST primary} exponent reflects how the QED running coupling responds to substrate-level Q-P anti-commutation. (Multi-week explicit derivation work; v0.1 establishes structural framing.)

2.3 Substrate-parity vs experimental-parity distinction

This finding sharpens an important conceptual point:

- Experimental parity P (3D spatial reflection): preserves Q in QED, violated in weak interactions
- **Substrate-level parity P̂_op** (T2472 Möbius + Pin(2) Z_2): anti-commutes with Q̂ structurally on Wallach K-types

The substrate-experimental projection map must therefore decompose P̂_op into: P̂_op = P_3D-parity (commutes with Q) ⊗ C_charge-flip (anti-commutes with Q)

at the experimental level. The projection of P̂_op onto experimental observables gives the standard P operator AND the C charge-conjugation in product form. Substrate-parity is structurally substrate-CP at the experimental level.

(Cal #27 STANDING reflexive trigger: this feels substrate-natural. Honest scope check — is this an INTERPRETATION of the {Q̂, P̂_op} = 0 finding, or a derivation? It's an interpretation that's consistent with the algebra but requires explicit projection-map analysis to confirm. Multi-week work; v0.1 INTERPRETIVE for this projection claim.)

2.4 Connection to 8-sided die hypothesis (A_sub v0.4 Section 6)

The 8 = 2^N_c potential commitment paths per substrate-tick include charge choices. With {Q̂, P̂_op} = 0:

- Of the 8 paths, ones with definite (charge sign, parity sign) come in anti-commuting pairs
- This forces structural constraints: only certain (Q, P) combinations are simultaneously diagonalizable
- The "1 unstable face" hypothesis (Section 6.3) could correspond to the (Q, P) combination that violates anti-commutation simultaneous-diagonalizability

This is one candidate substrate-mechanism for the unstable face in the 8-sided die. Other candidates (color-singlet violation, Mersenne-prefix forbidden, mixed-grading) remain possible per A_sub v0.4 Section 6.3 enumeration. Multi-week resolution.

3. Graph framing — edge structure from [Q̂, P̂_op]

Per A_sub v0.4 Section 3: A_sub generator commutators induce K-type graph edges. The [Q̂, P̂_op] = -2 Q̂ · P̂_op structure gives:

Edge type from Q̂: self-loops on each K-type (Q̂ is K-type-diagonal); edge weight = m_2 (charge eigenvalue).

Edge type from P̂_op: edges (m_1, m_2) → (m_1, -m_2) with weight ε_K · (-1)^{m_2}; these are σ-induced parity transitions.

Composite edge from [Q̂, P̂_op]: edges (m_1, m_2) → (m_1, -m_2) with weight -2 m_2 · ε_K · (-1)^{m_2}.

Interpretive observation: edges weighted by charge eigenvalue × parity sign × chirality grading. Higher-charge K-types have stronger parity-induced transitions. The substrate's reaction-table has charge-weighted parity transitions as part of its edge structure.

This is the first concrete piece of the A_sub v0.4 reaction-table edge enumeration.

4. Honest scope (Cal #27 STANDING)

What's STRUCTURALLY VERIFIED CANDIDATE in v0.1: - The {Q̂, P̂*op} = 0 anti-commutator computation is straightforward algebra given T2470 + T2472 definitions - The result follows directly from σ : V*(m_1, m_2) → (-1)^{m_2} V_(m_1, -m_2) which is standard Möbius-involution action - Promotion to STRUCTURALLY VERIFIED pending Cal cold-read of Section 1.2 algebra

What's INTERPRETIVE in v0.1: - Section 2.1 "substrate-parity = substrate-CP at experimental level" — claim consistent with algebra but requires explicit projection-map verification (multi-week) - Section 2.2 "α^{BST primary} reflects {Q̂, P̂_op} = 0 anti-commutation" — load-bearing but requires explicit QED loop-expansion analysis to derive (multi-week) - Section 2.4 "unstable die-face = (Q, P) anti-commutation simultaneous-diagonalizability violation" — one candidate among several (A_sub v0.4 Section 6.3)

Cal #27 STANDING reflexive trigger count: 1 reflexive trigger this doc (Section 2.3 substrate-parity-as-substrate-CP feels substrate-natural). Honest-scope checked — flagged as INTERPRETIVE pending projection-map analysis.

What's NOT in v0.1 (multi-week extensions): - Explicit QED loop-expansion derivation of α^{BST primary} from {Q̂, P̂_op} = 0 - Projection map P̂_op → P_3D-parity ⊗ C_charge-flip explicit decomposition - Resolution of which (if any) of the 4 candidate unstable-face mechanisms (A_sub v0.4 Section 6.3) is correct

5. Next commutator (A_sub v0.5 step 2 candidates)

Per A_sub v0.4 Section 3.3 + 12, the load-bearing commutators are:

1. □ [Q̂, P̂_op] — closed here, v0.1
2. [T̂_tick, Ĥ_sub] — substrate-tick evolution structure (load-bearing for SWPP dynamics)
3. [B̂, T̂_tick] — Bell-CHSH dynamical structure (load-bearing for Track DC)

4. [$\hat{\gamma}^5$, T̂] — chirality with time-reversal
5. [$\hat{\gamma}^5$, Ĉ] — chirality with charge-conjugation (CP-violation source)
6. [Ŝ_i, Ŝ_j] across sublattices
7. [L̂_i, $\hat{\gamma}^5$] — angular momentum with chirality
8. [B̂, Q̂] — Bell-CHSH with charge
9. [Ĉ_3, Î_3] — color with weak isospin (electroweak gauge non-commutativity)

Recommended next: [T̂_tick, Ĥ_sub] — substrate-tick evolution structure is the most novel and load-bearing for SWPP-language unification. After that: [B̂, T̂_tick] for Track DC's Bell 1/8 mechanism.

6. Coordination

Cal: cold-read of Section 1 algebra + tier-discipline check on Sections 2.1-2.4 interpretive content. Type C level-crossing per Cal #122 if {Q̂, P̂_op} = 0 supports Strong-Uniqueness criterion candidate (probably not — it's substrate-structural property derived from T2470 + T2472; doesn't advance Cal #99 META-theorem discipline).

Keeper: K-audit chain entry for [Q̂, P̂_op] = -2 Q̂ · P̂_op structural identity.

Elie: K52a Session 7+ multi-year work informs broader A_sub commutation structure; Toy 3531+ K-type population becomes more tractable as commutators close.

Grace: catalog entry for {Q̂, P̂_op} = 0 structural identity + 8-sided die hypothesis candidate mechanism.

— Lyra, A_sub commutator closure step 1 [Q̂, P̂_op] v0.1 filed Tuesday 2026-05-26 ~08:25 EDT per Task #322 v0.4 Section 3.3 work plan. STRUCTURALLY VERIFIED CANDIDATE pending Cal Section 1.2 algebra cold-read. Substantive finding: substrate-parity P̂_op anti-commutes with charge Q̂ — substrate-parity contains charge-flipping content via Möbius involution σ on SO(2) factor.